Biological Barriers and Pathways

1 Virtually all living things have some way of getting from here to there. Animals may walk, swim, or fly. Plants and their seeds drift on wind or water or are carried by animals. Therefore, it is reasonable to expect that, in time, all species might spread to every place on Earth where favorable conditions occur. Indeed, there *are* some cosmopolitan species. A good example is the housefly, found almost everywhere on Earth. However, such broad distribution is the rare exception. Just as barbed wire fences prevent cattle from leaving their pasture, biological barriers prevent the dispersal of many species.

2 What constitutes barriers depends on the species and its method of dispersal. Some are physical barriers. For land animals, bodies of water, chains of mountains, or deserts are effective. For example, the American bison spread throughout the open grasslands of North America, but in the southern part of the continent there are deserts, so the bison could not spread there. For aquatic creatures, strong currents, differences in salinity, or land areas may serve as barriers.

3 Some barriers involve competition with other species. A dandelion seed may be carried by the wind to bare ground, and, if environmental factors are right, it may germinate. There is not much chance, however, that any individual seedling will survive. Most places that are suitable for the growth of dandelions are already occupied by other types of plants that are well adapted to the area. The dandelion seedling must compete with these plants for space, water, light, and nutrients. Facing such stiff competition, the chances of survival are slim.

4 For animals, some barriers are behavioral. The blue spotted salamander lives only on mountain slopes in the southern Appalachian Highlands. Although these creatures could survive in the river valleys, they never venture there. Birds that fly long distances often remain in very limited areas. Kirkland's warblers are found only in a few places in Michigan in the summer and fly to the Bahamas in winter. No physical barriers restrict the warblers to these two locations, yet they never spread beyond these boundaries. Brazil's Amazon River serves as a northern or southern boundary for many species of birds. They could freely fly over the river, but they seldom do.

5 There are three types of natural pathways through which organisms can overcome barriers. One type is called a *corridor.* A corridor consists of a single type of habitat that passes through various other types of habitat. North America's Rocky Mountains, which stretch from Alaska to northern Mexico, is an example. Various types of trees, such as the Engelmann spruce, can be found not only at the northern end of the corridor in Alaska but also at higher elevations along the entire length of this corridor.

6 A second type of natural pathway is known as a *filter route.* A filter route consists of a series of habitats that are different from one another but are similar enough to permit organisms to gradually adapt to new conditions as they spread from habitat to habitat. The greatest difference between a corridor and a filter route is that a corridor consists of one type of habitat, while a filter consists of several similar types.

7 The third type of natural pathway is called a *sweepstakes route.* This is dispersal caused by the chance combination of favorable conditions. Bird watchers

are familiar with "accidentals," which are birds that appear in places far from their native areas. Sometimes they may find a habitat with favorable conditions and "colonize" it. Gardeners are familiar with "volunteers," cultivated plants that grow in their gardens although they never planted the seeds for these plants. Besides birds and plants, insects, fish, and mammals also colonize new areas. Sweepstakes routes are unlike either corridors or filter routes in that organisms that travel these routes would not be able to spend their entire lives in the habitats that they pass through.

8 Some organisms cross barriers with the intentional or unintentional help of humans, a process called *invasion.* An example is the New Zealand mud snail, which was accidentally brought to North America when trout from New Zealand were imported to a fish hatchery in the United States. It has caused extensive environmental damage in streams and rivers. In the invasive species' native environments, there are typically predators, parasites, and competitors that keep their numbers down, but in their new habitat, natural checks are left behind, giving the invaders an advantage over native species. Invasive species may spread so quickly that they threaten commercial, agricultural, or recreational activities.

Glossary

salamander: *a type of amphibian animal resembling a lizard*

sweepstakes: *a game of chance; a lottery*

1 of 26 The word cosmopolitan in the passage is closest in meaning to

○ worldwide
○ useful
○ well-known
○ ancient

2 of 26 How does the author explain the concept of biological barriers in paragraph 1?

○ By providing several examples of biological barriers
○ By describing the process by which barriers are formed
○ By comparing biological barriers with a familiar man-made barrier
○ By explaining how houseflies have been affected by biological barriers

3 of 26 What does the author suggest about American bison in paragraph 2?

○ They spread to North America from South America.
○ A body of water stopped them from spreading south.
○ They require open grasslands to survive.
○ They originally lived in deserts.

4 of 26 According to the passage, very few dandelion seedlings survive because of

○ the danger of strong winds
○ competition from other dandelions
○ the lack of a suitable habitat
○ competition from other species

5 of 26 In this passage, the author does NOT provide a specific example of

- ◯ a bird that is affected by behavioral barriers
- ◯ an aquatic animal that is blocked by physical barriers
- ◯ a land animal that is affected by behavioral barriers
- ◯ a tree that has spread by means of a corridor

6 of 26 The word slim in this passage is closest in meaning to

- ◯ unknown
- ◯ impossible
- ◯ remarkable
- ◯ unlikely

7 of 26 The phrase these two locations in paragraph 4 refers to

- ◯ Michigan and the Appalachian Highlands
- ◯ Brazil and the Bahamas
- ◯ the Appalachian Highlands and Brazil
- ◯ the Bahamas and a few places in Michigan

8 of 26 Why does the author mention the Amazon River in paragraph 4?

- ◯ To give an example of an important physical barrier
- ◯ To point out that many migrating birds fly across it
- ◯ To provide an example of a behavioral barrier
- ◯ To describe a barrier that affects aquatic animals

9 of 26 According to paragraph 6, how does the author distinguish a filter route from a corridor?

- ◯ A corridor consists of one habitat for its entire length, but a filter route consists of more than one.
- ◯ Organisms cannot live all of their lives in some parts of a filter route, but they can in a corridor.
- ◯ The distance from one end of a filter route to the other end is longer than the distance from one end of a corridor to the other.
- ◯ Plants spread through a corridor, while animals spread through a filter route.

10 of 26 In paragraph 8, the author gives New Zealand mud snails as an example of

- ◯ an invasive species that was unintentionally transported to another habitat
- ◯ a native species that has been damaged by an invasive species
- ◯ an invasive species that was intentionally brought to a new environment
- ◯ an animal that spread by means of a sweepstakes route

11 of 26 Which of the following sentences best expresses the essential information in the sentence below? (Incorrect answer choices omit important information or change the meaning of the original sentence in an important way.)

In the invasive species' native environments, there are typically predators, parasites, and competitors that keep their numbers down, but in their new habitat, natural checks are left behind, giving the invaders an advantage over native species.

○ Invasive species are organisms that leave their native environments behind and move to a new environment.
○ Native species are at a disadvantage compared to invasive species because they face environmental dangers that invasive species have left behind.
○ The greatest danger from invasive species is that they may spread parasites among native species.
○ In a new environment, predators, parasites, and competitors prevent invasive species from spreading faster than native species.

12 of 26 Look at the four squares [■] that indicate where the following sentence could be added to the passage.

They may be blown off course by storms or may be escaping population pressures in their home areas.

The third type of natural pathway is called a *sweepstakes route.* This is dispersal caused by the chance combination of favorable conditions. ■ Bird watchers are familiar with "accidentals," which are birds that appear in places far from their native areas. ■ Sometimes they may find a habitat with favorable conditions and "colonize" it.■ Gardeners are familiar with "volunteers," cultivated plants that grow in their gardens although they never planted the seeds for these plants. ■ Besides birds and plants, insects, fish, and mammals also colonize new areas. Sweepstakes routes are unlike either corridors or filter routes in that organisms that travel these routes would not be able to spend their entire lives in the habitats that they pass through.

Circle the square [■] *that indicates the best place to add the sentence.*

13 of 26 DIRECTIONS: Below is an introductory sentence for a brief summary of the passage. Complete the summary by writing the letters of **three** of the answer choices that express the most important ideas of the passage. Some of the answer choices are incorrect because they express ideas that are not given in the passage or because they express only details from the passage.

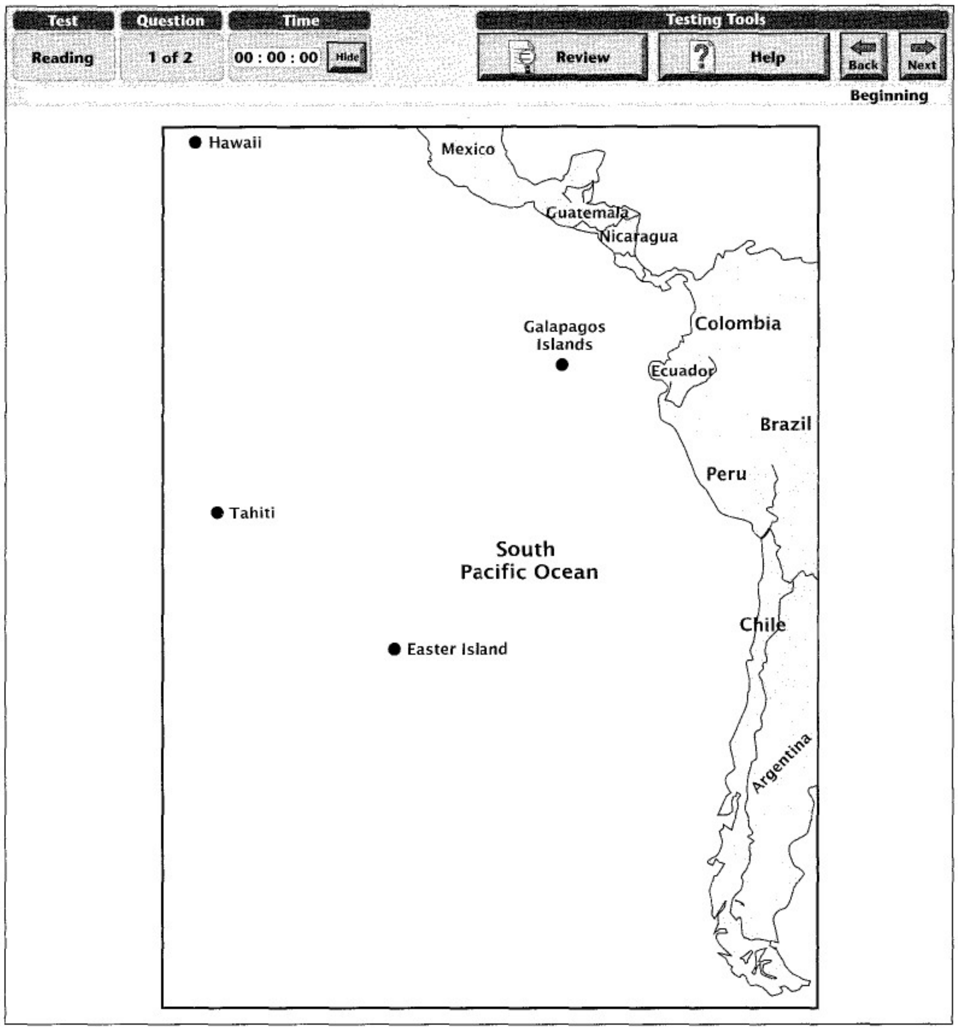
Biological barriers prevent organisms from spreading to all habitats where conditions are suitable.

- ____________________
- ____________________
- ____________________

Answer Choices

A. Organisms that spread by means of sweepstakes routes include species of birds called *accidentals* that appear in places far from their homes.
B. Biological barriers can be the result of physical features, climate, competition, and behavior.
C. Organisms can cross barriers by means of three types of natural pathways: corridors, filter routes, and sweepstakes routes.
D. Behavioral barriers do not prevent the spread of species from place to place as effectively as physical barriers.
E. Humans may accidentally or intentionally bring some species across natural barriers, and these species may have certain advantages over native species.
F. American bison spread throughout the grasslands of North America.

Mysteries of Easter Island



1 Easter Island is an isolated island in the Pacific between Chile and Tahiti. The island is roughly triangular and covers only 64 square miles (165 square kilometers). Because of its immense statues, Easter Island has long been the subject of curiosity.

2 There are 887 carved stone statues, called Moai, on Easter Island (not all complete). It is not known exactly what significance the Moai had to the Easter Islanders, but they were obsessed with building these statues. Some statues are as tall as 33 feet (11 meters) and weigh as much as 165 tons (167 metric tons). All portray a human head and sometimes an upper body. They are all carved from stone taken from a volcano on the island. Some are topped with a red "hat" called a *pukao,* made from a different type of stone, and a few have white coral eyes. The statues were moved on a network of roads on rollers made of palm logs and were then placed on stone bases called *ahu*. Most were built between 800 and 1500 A.D.

3 By the eighteenth century, the population had grown too large for the small island. At its peak, it was around 12,000. The only crop—sweet potatoes—could no longer feed the population. The palm forests had been cut down to provide rollers for the statues and to make way for roads. In 1722, when the first westerner, Admiral Jacob Roggeveen, visited the island, he wrote that there were hundreds of statues standing. When Captain Cook visited in 1774, he reported that only nine statues were still standing. Obviously, something dramatic had occurred during those years.

4 Any commentary about Easter Island would be incomplete without mentioning the theories of the Norwegian explorer and scientist Thor Heyerdahl, who came to the island in the 1950's. Heyerdahl learned that there had been two groups of islanders: the Hanau Momoko and Hanau Eepe—names once mistranslated as "Short Ears" and "Long Ears." The Hanau Momoko were dark-haired, the Hanau Eepe mostly red-haired. The Hanau Eepe used heavy earrings to extend the length of their ears. Heyerdahl theorized that the Hanau Momoko were Polynesians from other Pacific islands, but that the Hanau Eepe came later in rafts from South America. He believed that the Hanau Momoko became the servants of the Hanau Eepe, who forced them to build the statues. Because the Hanau Eepe were the masters, the statues resembled them. Heyerdahl said that the red "hats" of the statues actually represented the red hair of the Hanau Eepe. He also pointed out that the ears of the statues resembled those of the Hanau Eepe. According to Heyerdahl's theory, the Hanau Momoko eventually rose up in revolt, overturning most of the statues and killing off all but a few Hanau Eepe.

5 Heyerdahl gave other evidence for the South American origin of the Hanau Eepe. The stonework of the stone platforms called *ahu* was incredibly intricate, unlike any made by other Pacific Islanders. However, the Inca people of South America were famous for intricate stonework. Another piece of evidence Heyerdahl presented was the fact that the staple food of the Easter Islanders, the sweet potato, is not found in Polynesia. He believed that it came with the Hanau Eepe from South America.

6 DNA testing has proven that all Easter Islanders were in fact descended from Polynesians. The current theory is that the Hanau Momoko and Hanau Eepe were two of perhaps twelve clans of islanders, all of whom built statues. The "statue toppling wars" broke out among the clans as the island became overpopulated. When one group won a victory over another, they toppled their enemies' statues. Archaeologists say that the resemblance between the stonework of the Easter Islanders and that of the Inca is coincidental. As for the sweet potato, most scientists now believe that sweet potato seeds came to the island in the stomachs of sea birds.

7 Mysteries about the Moai of Easter Island certainly remain, but current archaeological research has made one lesson clear: overpopulation and overuse of resources such as occurred on Easter Island can lead to the downfall of thriving societies.

Glossary

clans: *social units larger than families but smaller than tribes*

toppling: *knocking over; overturning*

14 of 26 The word immense in paragraph 1 is closest in meaning to

- ◯ large
- ◯ strange
- ◯ ancient
- ◯ ruined

15 of 26 According to the information in paragraph 2, which of the following type of Moai is the LEAST common?

- ◯ Those that are carved from volcanic stone
- ◯ Those with red stone "hats"
- ◯ Those with white coral eyes
- ◯ Those that portray a human head

16 of 26 Which of the following best explains the term *ahu* in paragraph 2?

- ◯ Platforms made of stone
- ◯ Red stone "hats"
- ◯ Rollers made from palm logs
- ◯ Specially constructed roads

17 of 26 What does the author refer to with the phrase something dramatic in paragraph 3?

- ◯ The arrival of westerners
- ◯ The toppling of the statues
- ◯ The destruction of the forests
- ◯ The building of the statues

18 of 26 Which of these statements best reflects the author's opinion of the theories of Thor Heyerdahl?

- ◯ They are important but incorrect.
- ◯ They are strange but true.
- ◯ They are valid but incomplete.
- ◯ They are outdated but useful.

19 of 26 In paragraph 4, the author says that the terms *Hanau Momoko* and *Hanau Eepe*

- ◯ mean "Dark-Haired" and "Red-Haired" in the language of the Easter Islanders
- ◯ originally come from the language of the Inca
- ◯ have never been accurately translated into English
- ◯ do not really mean "Short Ears" and "Long Ears" in the language of the Easter Islanders

20 of 26 What can be inferred from the information in paragraph 4 about the ears of the Easter Island statues?

- ◯ They were broken off in the statue-toppling wars.
- ◯ They were not made of the same kind of stone as the other parts of the statues.
- ◯ They were long like those of the Hanau Eepe.
- ◯ They were not made of stone but of wood from palm trees.

21 of 26 The word intricate in paragraph 5 is closest in meaning to

- ○ heavy
- ○ complex
- ○ colorful
- ○ breakable

22 of 26 According to modern theory, how did sweet potato seeds come to Easter Island?

- ○ They were brought from South America.
- ○ They were washed up by the waves.
- ○ They were brought by westerners in 1722.
- ○ They were transported by sea birds.

23 of 26 The main point of paragraph 7 is to

- ○ argue that more research is needed
- ○ point out certain dangers that can destroy societies
- ○ summarize recent research
- ○ explain why some mysteries will never be solved

24 of 26 The word thriving in paragraph 7 is closest in meaning to

- ○ isolated
- ○ divided
- ○ successful
- ○ remarkable

25 of 26 Look at the four squares [■] that indicate where the following sentence could be added to the passage.

> **After all, they say, the statues themselves show that the islanders were skilled stone workers.**

DNA testing has proven that all Easter Islanders were in fact descended from Polynesians. ■ The current theory is that the Hanau Momoko and Hanau Eepe were two of perhaps twelve <u>clans</u> of islanders, all of whom built statues. ■ The "statue *toppling* wars" broke out among the clans as the island became over-populated. When one group won a victory over another, they toppled their enemies' statues. ■ Archaeologists say that the resemblance between the expert stonework of the Easter Islanders and that of the Inca is coincidental. ■ As for the sweet potato, most scientists believe that sweet potato seeds came to the island in the stomachs of sea birds.

Circle the square [■] that indicates the best place to add the sentence.